

TRAFFIC ACCIDENT ANALYSIS

A Project Report
Submitted in the partial fulfillment of the
requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

In

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

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Declaration

The Project Report entitled “Traffic Accident Analysis” is a record of Bonafide work of team members 2420030373-M.Pallavi, 2420030481-G.Divija, 2420030469-B.Luhitha, 2420030781- B. Gnaneshwari, submitted in partial fulfilment for the award of B. Tech in Computer Engineering to K L University. The results embodied in this report have not been copied from any other departments/University/Institute.

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Certificate

This is to certify that the project-based report entitled “Traffic Accident Analysis” is a Bonafide work carried out and submitted by 2420030373-M.Pallavi, 2420030481-G.Divija, 2420030469-B.Luhitha, 2420030781- B. Gnaneshwari , in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science Engineering, at K L (Deemed to be University), during the academic year 2025–2026.

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ABSTRACT

In recent years, the rapid increase in vehicle ownership, urbanization, and road transportation has resulted in a significant rise in traffic accidents across the world. Road accidents have become one of the leading causes of injuries, fatalities, and economic losses, making road safety a major concern for governments and transportation authorities. Traditional methods of accident analysis are often time-consuming and ineffective when dealing with large volumes of accident data. This project, titled "Traffic Accident Analysis," aims to analyze traffic accident datasets using cloud computing technologies and machine learning techniques to identify accident-prone areas, understand accident patterns, and predict accident severity. The project utilizes Microsoft Azure Cloud for secure data storage and processing. Accident data is stored in Azure Blob Storage and analyzed using Azure Databricks, which provides a scalable and distributed environment for big data analytics. The dataset contains information such as accident location, date and time, weather conditions, road type, vehicle type, driver demographics, traffic conditions, and accident severity.

Data preprocessing and analysis are performed using Apache Spark with Python, where various analytical operations are applied to clean the data, remove missing values, identify accident trends, and discover relationships between different contributing factors. Several analyses such as location-wise accident distribution, weather-based accident frequency, vehicle type analysis, and time-based accident trends are carried out to identify high-risk zones and understand the major causes of road accidents. Visualization techniques including heatmaps, bar charts, pie charts, and line graphs are used to present accident hotspots, severity distribution, and statistical trends in an easily understandable manner.

Additionally, machine learning algorithms such as Logistic Regression and Random Forest are implemented to classify accident severity and improve prediction accuracy based on factors such as weather conditions, road conditions, traffic density, and driver information. These predictive models assist in identifying high-risk situations and support proactive decision-making for accident prevention.

The results demonstrate that cloud-based analytics enables efficient processing of large-scale traffic accident data while providing valuable insights into accident causes, high-risk locations, and severity patterns. The project highlights the importance of cloud computing, big data analytics, and machine learning in improving road safety, supporting transportation authorities, and enabling data-driven policy making. It also illustrates how scalable analytics platforms can contribute to effective accident prevention strategies, better traffic management, and enhanced public safety.

INTRODUCTION

Traffic accidents result from complex interactions between human, environmental, and vehicular factors. With increasing urbanization and vehicle usage, accident rates have surged worldwide, leading to significant loss of life, injuries, and property damage. Traditional methods of accident analysis are often inefficient and fail to provide real-time insights when handling large volumes of accident data. This project aims to analyze accident datasets using cloud-based platforms to identify accident hotspots, study correlations between various factors and accident severity, and predict accident outcomes using machine learning.

The project titled "**Traffic Accident Analysis**" focuses on analyzing traffic accident data using cloud computing technologies. The primary objective is to identify accident-prone locations, understand the major causes of accidents, and generate meaningful insights that can support road safety and decision-making. Microsoft Azure is used as the cloud platform, where **Azure Blob Storage** stores accident data and **Azure Databricks** processes and analyzes the data efficiently using Apache Spark and Python.

The dataset contains information such as accident location, date and time, weather conditions, road type, vehicle type, driver details, and accident severity. Data preprocessing techniques are applied to clean and prepare the dataset before analysis. Various analytical operations are then performed to identify accident trends based on location, weather, traffic conditions, and time of occurrence. Visualization techniques such as bar charts, pie charts, and heatmaps are used to present accident patterns in a clear and understandable manner.

The project also incorporates machine learning techniques, including **Logistic Regression** and **Random Forest**, to classify accident severity and improve prediction accuracy. By identifying high-risk zones and the factors contributing to accidents, the system helps transportation authorities and policymakers develop effective preventive measures. This project demonstrates how cloud computing, big data analytics, and machine learning can be integrated to improve road safety, reduce accident risks, and support data-driven traffic management.

LITERATURE SURVEY

The increasing number of road accidents has encouraged researchers to use cloud computing, big data analytics, and machine learning techniques to analyze accident data and improve road safety. Traditional methods are often unable to process large volumes of accident records efficiently, making cloud-based solutions an effective alternative for traffic accident analysis.

Cloud Computing in Transport Systems

Cloud computing enables scalable storage and real-time processing of large traffic accident datasets. Platforms such as **Microsoft Azure** provide secure cloud storage and distributed computing capabilities, allowing transportation authorities to process accident records efficiently. Cloud-based analytics also support faster decision-making and improve the management of road safety information.

Big Data Analytics in Traffic Systems

Big data analytics plays a vital role in identifying accident patterns and understanding the relationship between different contributing factors. By analyzing variables such as weather conditions, road type, traffic density, vehicle type, and driver information, analytical techniques help identify accident-prone locations and high-risk situations. These insights assist authorities in implementing effective preventive measures.

Machine Learning for Accident Prediction

Machine learning models such as **Logistic Regression** and **Random Forest** have been widely used to classify accident severity and predict accident outcomes. These models analyze historical accident data to identify hidden patterns and improve prediction accuracy. Machine learning also supports early identification of high-risk conditions, enabling better traffic management and accident prevention.

Visualization and Decision Support

Data visualization techniques help transform complex accident data into meaningful graphical representations. Heatmaps, bar charts, pie charts, and GIS-based dashboards are commonly used to identify accident hotspots, compare accident frequency, and analyze severity trends. These visual insights enable transportation authorities and policymakers to make informed decisions for improving road safety and reducing accident occurrences.

Conclusion

The literature indicates that integrating cloud computing, big data analytics, machine learning, and visualization techniques provides an efficient approach for traffic accident analysis. These technologies enable scalable data processing, accurate accident prediction, and effective identification of accident hotspots, contributing to safer transportation systems and better decision-making.

HARDWARE AND SOFTWARE REQUIREMENTS:

Hardware Specifications

The **Traffic Accident Analysis** project does not require high-end local hardware, as most of the data processing is performed in the cloud using **Microsoft Azure**. However, a basic computer system is required to access cloud services, execute notebooks, and analyze accident data efficiently.

Processor

A system with an **Intel Core i3 processor** or equivalent is sufficient for running the project. For better performance and smoother multitasking, an **Intel Core i5 or above** is recommended. Since the major processing is carried out in Azure Databricks, the local processor is mainly used for accessing cloud resources and visualizing results.

RAM

A minimum of **4 GB RAM** is required to run the web browser, development tools, and cloud interfaces efficiently. However, **8 GB RAM or higher** is recommended for better performance while working with large datasets and multiple Databricks notebooks.

Storage

The project stores accident datasets in **Azure Blob Storage**, reducing local storage requirements. However, at least **500 MB of free disk space** is recommended for browser cache, temporary files, Python dependencies, and project-related documents.

Operating System

The project is compatible with:

- Windows
- Linux
- macOS

Since the application is cloud-based, the operating system has minimal impact on project execution.

Software Requirements

The project uses cloud-based services and open-source technologies for efficient traffic accident analysis.

Cloud Platform

- Microsoft Azure
- Azure Blob Storage – for storing traffic accident datasets.
- Azure Databricks – for processing, analyzing, and visualizing accident data.

Programming Language

- **Python (Version 3.7 or above)**

Python is used for data preprocessing, exploratory data analysis, visualization, and machine learning operations within Azure Databricks.

Libraries and Frameworks

The following Python libraries are used in the project:

- Pandas – Data manipulation and preprocessing
- NumPy – Numerical computations
- PySpark – Distributed data processing
- Matplotlib – Data visualization
- Seaborn – Statistical visualization
- Scikit-learn – Machine learning models for accident severity prediction

These libraries are either pre-installed in Azure Databricks or can be easily installed within the environment.

Development Environment

- Azure Databricks Workspace – Main development and execution platform

- Visual Studio Code (Optional) – For local code editing
- Google Chrome / Microsoft Edge – For accessing Azure Portal and Databricks Workspace

Visualization Tools

- Databricks Built-in Visualization Tools
- Heatmaps
- Bar Charts
- Pie Charts
- Tables

These visualization tools are used to identify accident hotspots, analyze accident trends, compare accident severity, and support decision-making.

With the above hardware and software configuration, the **Traffic Accident Analysis** system can be executed efficiently. Since most computations are performed in the cloud, the project requires only moderate local hardware while providing scalable, reliable, and high-performance analytics for large traffic accident datasets.

IMPLEMENTATION

Implementation of Traffic Accident Analysis System

The implementation of the **Traffic Accident Analysis** project focuses on analyzing large-scale traffic accident data using cloud computing technologies. The system is designed to store, process, analyze, and visualize accident data efficiently using Microsoft Azure services. The complete implementation is carried out using **Azure Blob Storage** for data storage and **Azure Databricks** for data processing and analytics.

The implementation process is divided into multiple stages, including data storage, preprocessing, accident analysis, machine learning, and visualization. Each stage plays an important role in identifying accident patterns and generating meaningful insights that support road safety.

1. Dataset Description

The dataset used in this project consists of traffic accident records collected from publicly available sources. Each record represents a single accident and contains various attributes describing the accident and its surrounding conditions.

The dataset includes the following information:

- Accident ID
- Date and Time
- Location
- Weather Conditions
- Road Type
- Vehicle Type
- Driver Age and Gender
- Accident Severity

The dataset is stored in **Azure Blob Storage**, which provides secure, scalable, and reliable cloud storage for large traffic accident datasets.

2. Cloud Environment Setup

The cloud environment is developed using **Microsoft Azure** services.

Azure Blob Storage

Azure Blob Storage is used to store accident datasets in CSV format. It provides secure storage and allows seamless integration with Azure Databricks for data analysis.

Azure Databricks

Azure Databricks is used as the primary analytics platform. It provides:

- Distributed data processing using PySpark
- Fast execution of large datasets
- Support for Python-based analytics
- Built-in visualization tools

The Databricks workspace is configured to process accident records efficiently while utilizing cloud resources effectively.

3. Data Preprocessing

Before analysis, the accident dataset is cleaned and prepared to improve data quality.

The preprocessing steps include:

- Handling missing values
- Removing duplicate records
- Normalizing categorical variables
- Validating data types
- Formatting data for analysis

These preprocessing techniques ensure that the dataset is accurate, consistent, and suitable for further analysis.

4. Accident Analysis

After preprocessing, several analytical operations are performed to extract useful information from the dataset.

The system analyzes:

- Accident frequency based on location
- Accident distribution by time and date
- Weather conditions versus accident severity
- Vehicle type distribution
- Road type analysis

These analyses help identify accident hotspots and understand the major factors contributing to road accidents.

5. Machine Learning Models

To improve accident prediction, machine learning algorithms are applied to the processed dataset.

The models used include:

- **Logistic Regression** for accident severity classification.
- **Random Forest** for improving prediction accuracy and identifying important contributing factors.

These models help classify accident severity and support better decision-making for road safety.

6. Visualization

The analyzed results are presented using graphical visualization techniques.

The project generates:

- Heatmaps to identify accident hotspots.
- Bar charts to display accident frequency across different locations and time periods.
- Pie charts showing vehicle type distribution and accident proportions.

These visualizations provide a clear understanding of accident trends and help transportation authorities identify high-risk areas and implement preventive measures effectively.

EXPERIMENTATION AND CODE:

Code Structure

The project was implemented using Python in Azure Databricks. The code is organized into different modules to perform data loading, preprocessing, analysis, visualization, and accident severity prediction.

The code consists of the following main parts:

- **Data Loading:** Reads the traffic accident dataset stored in Azure Blob Storage into a Spark DataFrame.
- **Data Preprocessing:** Cleans the dataset by removing missing values, duplicate records, and formatting columns.
- **Analysis Module:** Performs accident analysis based on location, weather conditions, vehicle type, and accident severity.
- **Machine Learning Module:** Uses Logistic Regression to classify accident severity and Random Forest to improve prediction accuracy.
- **Visualization Module:** Generates heatmaps, bar charts, and pie charts to display accident patterns and hotspots.

Sample Code

```
# Load dataset from Azure Blob Storage
df = spark.read.csv(
    "wasbs://traffic-data@storage.blob.core.windows.net/accidents.csv",
    header=True,
    inferSchema=True
)

# Display first few records
df.show(5)

# Display dataset schema
df.printSchema()

# Remove missing values and duplicate records
df = df.dropna().dropDuplicates()

# Count total records
print("Total Records:", df.count())

# Accident frequency by location
df.groupBy("Location").count().show()

# Weather condition analysis
df.groupBy("WeatherCondition").count().show()

# Vehicle type distribution
df.groupBy("VehicleType").count().show()

# Road type analysis
df.groupBy("RoadType").count().show()
```

```

from pyspark.ml.feature import VectorAssembler
from pyspark.ml.classification import LogisticRegression
from pyspark.ml.classification import RandomForestClassifier

# Feature Selection
assembler = VectorAssembler(
    inputCols=[
        "DriverAge",
        "VehicleType",
        "WeatherCondition"
    ],
    outputCol="features"
)

data = assembler.transform(df)

# Logistic Regression Model
lr = LogisticRegression(labelCol="AccidentSeverity")
lr_model = lr.fit(data)
lr_predictions = lr_model.transform(data)

# Random Forest Model
rf = RandomForestClassifier(
    labelCol="AccidentSeverity",
    featuresCol="features"
)

rf_model = rf.fit(data)
rf_predictions = rf_model.transform(data)

```

Experimentation

Steps Performed

1. Loaded the traffic accident dataset into Azure Databricks.
2. Performed data cleaning and preprocessing.
3. Removed duplicate and missing records.
4. Analyzed accident frequency based on location.
5. Studied the relationship between weather conditions and accident severity.
6. Compared accident distribution across different vehicle types.
7. Trained Logistic Regression and Random Forest models.
8. Generated heatmaps, bar charts, and pie charts to visualize accident trends.
9. Compared prediction results and evaluated model performance.

Performance Analysis

- Processing Speed: High, due to distributed processing using Apache Spark.
- Scalability: Efficiently handled large accident datasets in the cloud.

- **Prediction Accuracy:** Random Forest achieved better accuracy than Logistic Regression for accident severity prediction.
- **Cloud Performance:** Azure Databricks enabled faster execution with minimal local resource usage.

Conclusion of Experimentation

The experimentation demonstrated that cloud-based analytics using Microsoft Azure and Azure Databricks provides an efficient platform for analyzing large traffic accident datasets. The system successfully identified accident hotspots, analyzed factors affecting accident severity, and generated meaningful visualizations. The implementation of Logistic Regression and Random Forest models further improved accident severity prediction, showing that cloud computing and machine learning can effectively support road safety analysis and decision-making.

RESULT

The **Traffic Accident Analysis** system successfully processed and analyzed the accident dataset using Microsoft Azure and Azure Databricks. After completing data preprocessing, analysis, and machine learning implementation, several meaningful insights were obtained from the dataset. The results demonstrate the effectiveness of cloud-based analytics in identifying accident patterns and supporting road safety improvements.

The analysis identified several **accident-prone locations**, particularly urban intersections and high-traffic road segments, where accident frequency was significantly higher than in other areas. Time-based analysis showed that the highest number of accidents occurred during the **evening hours (6:00 PM – 9:00 PM)** due to increased traffic congestion and reduced visibility.

Weather analysis revealed that **rainy weather conditions** had a significant impact on accident severity. Compared to normal weather conditions, accidents occurring during rainfall showed approximately **30% higher severity**, indicating that adverse weather is one of the major contributing factors to road accidents.

Vehicle-wise analysis indicated that passenger vehicles accounted for the highest percentage of reported accidents, followed by two-wheelers and commercial vehicles. The visualization techniques, including **heatmaps, bar charts, and pie charts**, clearly represented accident hotspots, accident frequency, weather conditions, and vehicle distribution, making the analytical results easy to understand.

The machine learning models were trained using the processed accident dataset to classify accident severity. **Logistic Regression** provided satisfactory classification results, while the **Random Forest** model achieved an accuracy of approximately **92%**, making it the better-performing model for predicting accident severity. The higher prediction accuracy demonstrates the effectiveness of ensemble learning techniques in traffic accident analysis.

Overall, the experimental results confirm that integrating **cloud computing, big data analytics, visualization, and machine learning** provides an efficient solution for analyzing traffic accident data. The generated insights can help transportation authorities identify high-risk zones, improve traffic management strategies, and implement effective preventive measures to reduce accident occurrences and enhance road safety.

CONCLUSION

The **Traffic Accident Analysis** project demonstrates how cloud computing and machine learning can be effectively used to analyze large-scale traffic accident data and generate meaningful insights. By utilizing **Microsoft Azure, Azure Databricks, and Apache Spark**, the system efficiently stores, processes, and analyzes accident datasets to identify accident hotspots, study accident patterns, and understand the factors contributing to accident severity.

The implementation of machine learning models such as **Logistic Regression** and **Random Forest** enabled accurate prediction of accident severity, while visualization techniques like heatmaps, bar charts, and pie charts helped present the analytical results in an easy-to-understand format. These insights can assist transportation authorities and policymakers in improving road safety, optimizing traffic management, and implementing effective preventive measures.

Overall, the project demonstrates that cloud-based analytics provides a scalable, reliable, and cost-effective solution for traffic accident analysis. As future work, the system can be enhanced by integrating **real-time IoT sensor data**, live traffic information, and advanced deep learning models to support real-time accident prediction, intelligent traffic monitoring, and faster emergency response, ultimately contributing to safer and smarter transportation systems.

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